may be desirable or not because the two operations likely have different mechanisms and may be better tracked separately.

- Probabilities: This metric measures the probability of the right answer, or the difference in probabilities of different answers. The main issue with such metrics is
  - Probabilities are non-linear, in the sense that they track the logits exponentially. For example, a model component adding +2 to a given logit can create a 1 or 40 percentage point probability increase, just depending on what the baseline was.
    As an example of the non-linearity consider the orange line in the figure above: A modest increase in logit difference around layer 17 converts to a jump in probability.
  - Probabilities also inherit the problems of the logprob metric, namely saturation and unspecificity.
    The figure shows the saturation effect for the orange line at layer >18.
- Binary and discrete metrics (Accuracy / top-k performance / rank / etc): These metrics round off each input to a discrete metric (and then tend to average over a bunch of inputs).
  - The problem with these is that generally many components contribute to a model's performance, with no single decisive contributor. Discrete metrics may suggest that some significant contributors are unimportant, because they aren't enough to cross a threshold. Alternatively, these metrics may suggest that one contributor among many is *all* that matters because it happens to be the one that pushes the model over the threshold. We generally recommend using continuous metrics instead.
    As an example consider the Figure above: The rank-based metric (red line) jumps around layer 15 when the corresponding logit passes the rank 1 and 0 thresholds, while it is not sensitive to any of the other changes.
  - Discrete metrics can be a good fit for confirmatory patching rather than exploratory patching, as in some sense accuracy *is* the metric we care about - can the model get the question right or not?
- Logits: We could just take the answer *logit* as a metric. This is somewhat unprincipled because logits have an arbitrary baseline (adding +1 to all logits would not affect the output) but tend to work in practice. Logit(John) often matches Logprob(John) without being affected by the downsides of the logprob metric.
  This metric can incorrectly pick up on components that just contribute to many logits. Ensuring that the residual stream and logits have mean zero (default in TransformerLens) can help address this.

Desired property:	Continuous (not discrete)	Track model confidence linearly	Measure John vs Mary independent of P(name)	False-positive when "breaking the model"
Logit Difference:	Yes	Yes	Yes	Yes
Logit:	Yes	Yes (typically)	Yes	Maybe
Logprob:	Yes	Not close to p=1	No	No
Probability:	Yes	No	No	No
Binary metrics:	No	No	No	No

5 Summary

In most situations, use activation patching instead of ablations. Different corrupted prompts give you different information, be careful about what you choose and try to test a range of prompts.

There are two different directions you can patch in: denoising and noising. These are *not* symmetric. Be aware of what a patching result implies!

- Denoising (a clean $\rightarrow$ corrupt patch) shows whether the patched activations were **sufficient to restore** the model behaviour. This implies the components make up a cross-section of the circuit.
- Noising (a corrupt $\rightarrow$ clean patch) shows whether the patched activations were **necessary to maintain** the model behaviour. This implies the components are part of the circuit.

Be careful when using metrics that are (i) discrete, (ii) overly sharp, or (iii) sensitive to unintended information. Ideally use a range of metrics, and try to have at least one metric that is continuous and roughly linear in logits such as logit difference or logprob. We recommend representing patching results in a big dataframe with a column per metric and row per patching experiment, and making a bunch of plots from this.

- Model top-k accuracy is discrete and can overrepresent changes at thresholds and shows no change for large effects that don't cross thresholds.
- Most effects from patching are linear and additive in logit space. Probability is exponential in logit space, so it overemphasises effects near a threshold and suppresses effects elsewhere, creating overly sharp patching plots
- Logprob can saturate, and cannot control for a patch that boosts both the correct and incorrect answer(s)

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